

Partial Shading of Solar Panels: Theoretical Analysis; MATLAB Modeling; and Remedies

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Abstract—The effects of partial shading can be severe, even when only a small fraction of the panel is shaded. This phenomenon can lead to several issues. **Reduced Power Output:** As mentioned earlier, solar cells are typically connected in series, meaning that the current flowing through the entire string is limited by the lowest-performing cell. A shaded cell acts as a bottleneck, reducing the current in the entire panel and thereby decreasing its output power. In extreme cases, the shaded cell can even reverse current flow, leading to energy loss. **Hot Spots and Damage:** Partial shading can create "hot spots" on the panel, where the shaded cells overheat due to the backflow of current. These hot spots can cause permanent damage to the solar panel, significantly reducing its lifespan. Prolonged exposure to these conditions may cause the affected cells to fail entirely, which can be costly to replace.

Keywords— *Solar Energy, Photo voltaic, Partial Shading, Simscape, MATLAB, Modeling.*

I. INTRODUCTION (HEADING I)

Solar energy has experienced exponential growth over the past decade, driven by the decreasing cost of manufacturing solar panels the market of photovoltaic technology is rapidly evolving with a Compound Annual Growth Rate (CAGR) equal to 34% between 2010 and 2020.[1]

Modern batteries with high charge-to-mass density have made it easier to store solar energy, thereby enhancing its viability as a consistent power source. This combination of affordability and efficiency has propelled solar panels to the forefront as one of the most popular and rapidly expanding sources of renewable energy worldwide [2].

The global push for sustainability, spearheaded by initiatives like the United Nations' 17 Sustainable Development Goals (SDGs), has positioned solar power as a key alternative to fossil fuels. Solar energy contributes directly to reducing carbon footprints, combating climate change, and promoting clean energy accessibility. As governments, organizations, and individuals transition to greener energy solutions, solar panels have become a cornerstone in the fight against environmental degradation.

Despite its advantages, solar photovoltaic (PV) technology is not without limitations. The efficiency of solar panel, currently, is about 30%.[3] One major challenge that impacts the already low energy efficiency of PV systems is partial shading. This phenomenon occurs when objects such as trees, buildings, or even debris block sunlight from reaching portions of the solar panel. While it may seem minor, partial shading can significantly reduce a solar panel's overall performance.

Partial shading disrupts the uniform exposure of sunlight across the surface of a solar panel, leading to a drop in power

output. Solar panels are typically composed of multiple interconnected cells that work together to generate electricity. When shading occurs, even on a small part of the panel, the affected cells produce less energy or, in some cases, none. This imbalance can create a bottleneck, reducing the efficiency of the entire system.

To address the challenges posed by partial shading, this study delves into the theoretical aspects of the issue. Through MATLAB modeling, the complex interactions within PV systems under shaded conditions can be simulated and analyzed. These simulations help in understanding the performance drop caused by partial shading and in identifying potential mitigation strategies.

A. What is Partial Shading?

Partial shading happens when a part of a solar panel or a solar array is obstructed from receiving direct sunlight. This obstruction can arise from numerous causes, including the presence of nearby trees casting passing clouds. Additionally, environmental debris such as dirt, fallen leaves from trees, or bird nests on the panel surface can also lead to partial shading shadows, tall buildings that block sunlight during certain times of the day, utility poles positioned close to the installation, or even transient factors like.

Unlike complete shading, which occurs when an entire panel is fully covered and unable to generate any electricity, partial shading affects only a specific section of the panel. However, the consequences of partial shading are often disproportionately significant [4, 5, 6]. This is due to the interconnected nature of solar cells within a panel. Solar cells are typically arranged in series or parallel configurations, meaning that the performance of one shaded cell can influence the output of the entire string or even the panel. This can result in reduced efficiency and lower energy output, potentially affecting the overall performance of the solar energy system. Advanced solutions like bypass diodes and microinverters are sometimes employed to mitigate these effects, but understanding and managing the causes of partial shading remains crucial for optimizing solar panel efficiency.

Most solar panels consist of multiple solar cells connected in series. When one cell is shaded, it creates a "bottleneck" that can affect the overall current generated by the panel.[7] This happens because the shaded cell limits the flow of electricity through the entire string of cells in the panel. As a result, the panel's overall efficiency is reduced, even if only a small section is shaded. Partial shading can create "hot spots" on the panel, where the shaded cells overheat due to the backflow of current. These hot spots can cause permanent damage to the solar panel, significantly reducing its lifespan. Prolonged exposure to these conditions may cause the affected cells to fail entirely, which can be costly to replace.

This study shows that when a cell in a panel is shaded, even if it is not a full shading, a negative voltage appears across the shaded cell which could risk a permanent damage to the cell. The study shows that shading of 50% of a single cell in the panel can cause a drop of around 70% in the overall panel output power. For a panel of 72 cells, this means an overall shading of only 0.69% yet causes a power drop of about 70%, it is a too highly cost. The intermittence solar irradiance due to alternating day and night and/or weather condition have the obvious limitations on the availability of solar energy, partial shading can only aggravate the situation further. Partial shading causes a negative impacts during the abundance period of the irradiance which, usually, suffer relative scarcity, attributable to geographical and seasonal factors.

The study shows a tremendous agreement between the theoretical analysis and the Simscape model of the problem. This proves that simscape has the knack for mimicking real life physical systems to great extent.

B. Mitigating the Effects of Partial Shading

Several techniques can be employed to minimize the adverse effects of partial shading on solar panels:

1. Bypass Diodes: These are integrated into solar panels to allow electricity to flow around shaded cells, reducing energy loss.
2. Maximum Power Point Tracking (MPPT): Advanced algorithms in solar inverters continuously adjust the operating point of the solar array to maximize energy output, even under shaded conditions.[9, 10]
3. Panel Orientation and Placement: Proper installation of solar panels, considering potential shading sources, can prevent significant energy losses.
4. Use of Microinverters or Power Optimizers: These devices work at the individual panel level to ensure that each panel operates at its maximum potential, independent of the performance of other panels in the system.

II. THEORETICAL ANALYSIS

Solar cells are simply diodes with large surface areas, with the Cathode side which is the n-type semiconductor exposed to sun; hence it is the upper side. The Anode (p-type semiconductor) which the lower side of the panel is mounted on metal sheets to dissipated heat. The solar cells can be thought of as photon-to-electron converters where the incident solar irradiance energizes the atoms in the cathode region which in turn gives rise to more electrons in the conduction band that are ready to flow through external load and convey electrical energy to it as shown in the schematic diagram of Figure 1.

Solar panels are composed of many individual cells that are interconnected in both series and parallel configurations. This arrangement allows the panels to efficiently convert sunlight into electrical energy, which is then supplied to the systems or installations they are designed to power.[11] The

series connection increases the voltage output, while the parallel configuration ensures a higher current, together optimizing the overall performance and energy delivery of the solar panel.

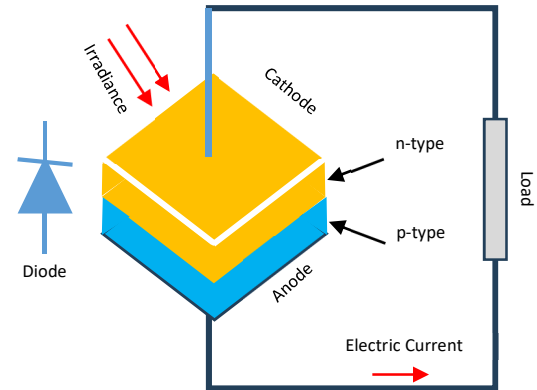


Figure 1 Schematic Diagram of a Solar Cell

This study concentrates on a panel of a single string of series-connected solar cells to investigate the stark effect of the partial shading. These results can then be generalized on other panels that have different arrangements of cells layout.

Figure 2 shows the equivalent circuit of a solar cell that behaves like a battery of around 0.6 Volt. [7, 8] It consists of four basic components; a current source; an ideal diode; a shunt resistance; and a parallel resistance. To simplify the analysis both resistances are going to be neglected. This approximation is not having a great deal of effect as the simulation results reveals, refer to the Simscape simulation section.

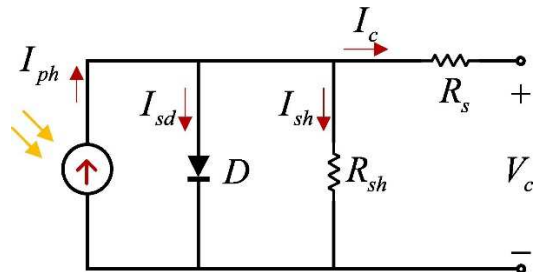


Figure 2 PV Cell Equivalent Circuit [3]

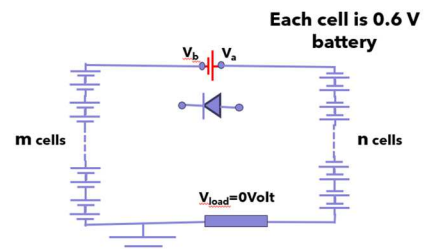


Figure 3 Circuit Diagram of a Solar Panel with a Shaded Cell

Figure 3 shows a 72-cell single string panel where all the cells are connected in series. Each cell is denoted by a battery of 0.6 Volt apart from the red-coloured cell which is presumed to be shaded while the rest are exposed to irradiance of 1000 W/m². The current source which is the photon to electron

converter I_{ph} of the shaded cell becomes zero; in other words, it becomes open circuit which leads to having this cell behaving just like an ordinary diode which in turn cause a blockage of the current running through the circuit. Applying Kirchhoff's law:

$$V_{ab} = V_a - V_b \quad (1)$$

$$V_{ab} = V_a - V_b \quad (2)$$

$$V_a = -0.6n \quad (3)$$

$$V_b = 0.6m \quad (4)$$

$$V_{ab} = -0.6(n + m) \quad (5)$$

Where:

$n + m = \text{total number of cells in the panel} - 1$

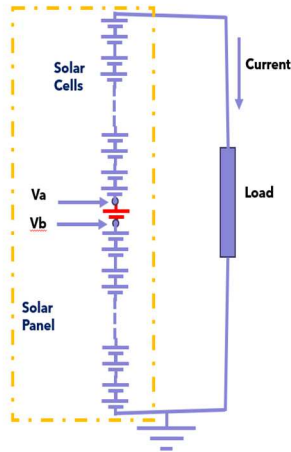


Figure 4 Schematic Diagram of Solar Panel interiors

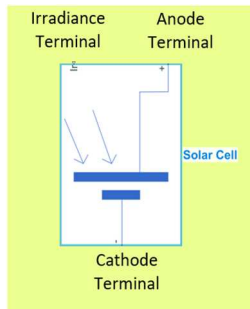


Figure 5 PV Cell Block in Simscape

The table below shows the voltage across the shaded cell for various panel sizes of PV panels.

Table 1 Shaded Cell Voltage V_{ab} for Different Panel Sizes.

Number of Cell in the Panel	Voltage of the Shaded Cell V_{ab} (V)
48	-28
72	-42
144	-86

III. Solar Panel Modeling

MATLAB modeling has been devised using Simulink and Simscape where 72 Photovoltaic Cells are cascaded in a single string to simulate a panel. Figure 5 represents the block diagram of the Simscape Model. Subsystem 1 and Subsystem 2 each consists of 35 PV cells. Figure 6 illustrates the Simscape PV cell. It shows three terminals: the Anode; the Cathode; and the Irradiant terminals. The Anode terminal is the positive battery side of the equivalent circuit, whereas the Cathode is the negative battery side of the equivalent circuit. The irradiant terminal, as the name suggests, is for simulating the irradiant input to the cell.[11] Simscape is a physical component level simulator whereas the Simulink is a block diagram level simulator package. MATLAB allows mix simulation of both if proper conversion blocks are used to match the signals of both worlds.

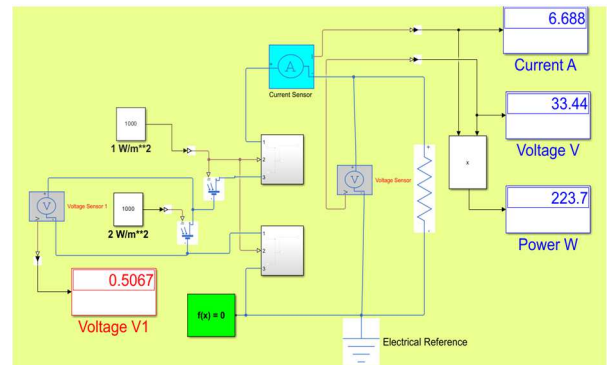


Figure 6 Block Diagram of Simscape Modeling of PV Panel

Table 3 shows the effect of shading on the performance of the panel; output power and the hotspot of the shaded PV cell expressed in terms of voltage across this cell.[7,8] At 50% shading the panel loses two third of its power and the created hotspot zone (shaded cell area) experiences negative 18.91 Volt. As the percentage shading increases further the performance of the panel degrades sharply to zero output power together with a big voltage drop on the shaded cell driving it further into reverse mode with high potential risk of permanent damage to the panel. The graphs of Figure 10 Illustrates this phenomenon.

Table 3 Effect of Percentage Partial shading of a Single Cell on Overall Power Output and Shaded Cell Voltage

#	Percentage Irradiance (%)	Solar Panel Output Power (W)	Voltage Across the Shaded Cell (V)	Remarks
0	0	223,70	0.50	normal operation, no shading
1	10	218.20	-0.20	
2	20	172.40	-5.60	
3	30	132.00	-10.29	
4	40	97.00	-14.68	
5	50	67.34	-18.91	power drops by 2 thirds
6	60	43.10	-23.04	
7	70	24.24	-27.10	

8	80	10.78	-31.10	
9	90	2.69	-35.07	
10	100	0.00	-39.00	zero watt output

Figure 7 is the graph of the data of table 3 and shows a knee in both curves between the 10% and 20% shading reading, this is due to the beginning of the blockage behavior of the shaded cell to the electrical current as this cell's diode shifts towards reverse bias and becomes more dominant. The voltage drop of the shaded cell drops linearly towards negative 40 point after the knee point which happens at around 15% shading. The output power curve of figure 7 starts a sharp downslope fall after the 15% shading. At 50% shading the panel loses more than two third of its power. The crippling effect on the performance of the panel is clearly demonstrated by this modeling which verifies and validates the theoretical analysis of this problem in section II.

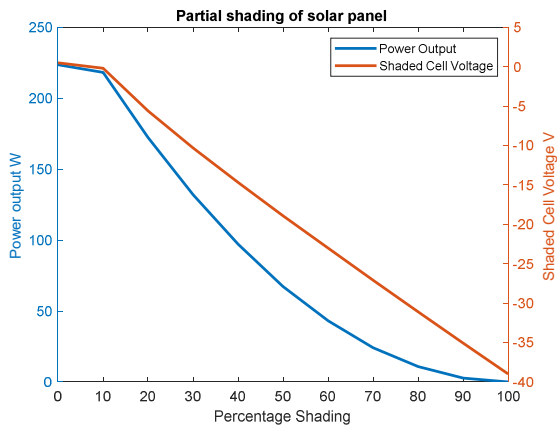


Figure 7 Effect of Shading on Panel Power Output and Shaded Cell Voltage

The percentage cell surface area of a 72 cell panel is only $1/72$ which is 0.0139 or 1.39% of total area of the panel. Further multiply this number 0.5 in case of 50% shading produces ($0.5 \times 1.39\% = 0.69\%$), this shows that tiny 0.69% of shading of the panel has degraded the output to a less one third of its normal operational output. This exhibits the seriousness and the impact of this problem on users of PV panels.

IV. Effect of Bypass Diode

Figure 8 shows the Simscape modeling of the partial shading of a single cell in a panel of 72 cell cascaded in a mono string. As it is clear from this figure a bypass diode has been added to model in parallel with the shaded cell oriented in the direction of the flow of the panel's current. This diode has provided an alternative route for the panel current surmounting the shaded-cell diode reverse mode. Despite the complete shading, the figure shows that an output of 199.4 W can be harnessed from this panel when a bypass diode is embraced in the circuit. This represents 90% of the regular power of the panel. Not all panel manufacturers deploy bypass diode due to cost and fabrication problems and space restrictions.

Surely, the best solution to partial shading is to eliminate the original cause of the problem whenever possible by tackling the obstacle rather than treating the symptoms. The study has shown that the partial has immense impact on the panel's yield.

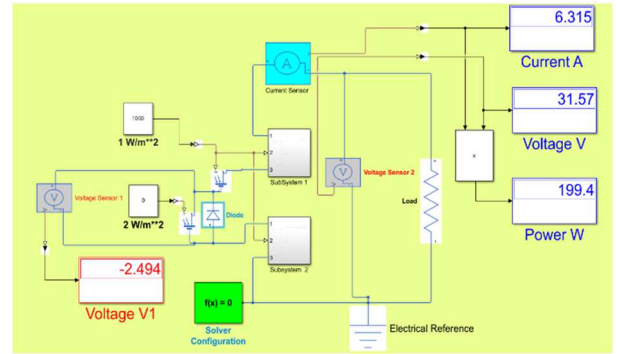


Figure 8 MATLAB Modeling of Simscape for different values of Shading

V. Conclusion, Limitations, and Future Directions

This study has provided a comprehensive analytical analysis of a single-string photovoltaic (PV) panel affected by partial shading, supported by detailed MATLAB modeling. The findings demonstrate a significant bottleneck effect caused by shaded cells, which severely restricts the panel's power output. In cases of full shading, the panel's output degrades entirely to zero, rendering it obsolete. Equation (5) from the analytical analysis and the results from the MATLAB simulation in Figure 11 vividly illustrate the critical impact of partial shading on solar panel efficiency.

The study has highlighted the disproportionately negative effects of shading, which accelerate the reduction in the output power of a solar panel. This can often result in the entire system stagnating and ceasing to function. Furthermore, the development of hotspots within shaded cells can cause permanent damage to the affected panel, necessitating costly replacements. These findings underscore the importance of addressing partial shading to enhance the reliability and lifespan of solar energy systems.

A. Limitations

One notable limitation of this study is its theoretical and simulated nature. While the analysis and MATLAB modeling provide valuable insights, the study lacks validation through practical implementation on physical PV systems. However, this work is rooted in real-world experiences where the author encountered the partial shading problem firsthand, which motivated the exploration of this topic in depth.

B. Future Work

Future research will focus on developing advanced optimization techniques to mitigate the effects of partial shading. A key direction involves integrating embedded microcontroller systems into solar panels. These systems would continuously scan individual cells within the panel and, with the assistance of Artificial Intelligence (AI), identify problematic cells affected by shading or damage.

The proposed solution includes the implementation of bypass mechanisms that dynamically isolate problematic cells to nullify their negative effects on the overall system. AI-driven algorithms could optimize the detection and bypassing

process, ensuring maximum efficiency and minimal energy loss under varying environmental and load conditions.

Additionally, future work could explore hybrid approaches that combine hardware innovations with software-based enhancements, such as predictive shading models and machine learning algorithms, to adaptively manage partial shading in real-time. Such advancements have the potential to revolutionize PV system performance and make solar energy more resilient to environmental challenges.

In conclusion, while this study offers a critical understanding of the partial shading problem, its true potential lies in inspiring practical solutions and innovations that will enable solar energy systems to achieve greater reliability and efficiency in the future.

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